anterior segments of the earthworm. They produce blood cells and haemoglobin which is dissolved in the plasma and gives red colour to the blood.

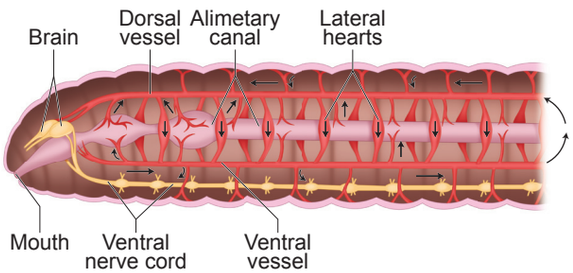


Figure 4.4 *Lampito mauritii*: Circulatory system and Nervous System

Nervous System

The bilobed mass of nervous tissue called supra - pharyngeal ganglia, lies on the dorsal wall of the pharynx in the 3rd segment, is referred as the "brain". The ganglion found below the pharynx in the 4th segment is called the sub-pharyngeal ganglion (Figure. 4.4). The brain and the sub - pharyngeal ganglia are connected by a pair of circum-pharyngeal connectives. They run one on each side of the pharynx. Thus a nerve ring is formed around the anterior region of the alimentary canal. The double ventral nerve cord runs backward from the sub - pharyngeal ganglion. The brain along with other nerves in the ring integrates sensory inputs and command muscular responses of the body.

The earthworm's receptors are stimulated by a group of slender columnar cells connected with nerves. The **Photoreceptors** (sense of light) are found on the dorsal surface of the body. **Gustatory** (sense of taste) and **olfactory receptors** (sense of smell) are found in the buccal cavity**. Tactile receptors** (sense of touch), **chemoreceptors** (detect chemical changes) and **thermoreceptors** (changes in temperature) are present in the prostomium and the body wall.

How do the earthworm's sense activity in their habitat without eyes, ears or a nose?

Excretory System

Excretion is the process of elimination of metabolic waste products from the body. In earthworm, excretion is effected by segmentally arranged, minute coiled, paired tubules called nephridia. There are three types of nephridia; (i) **pharyngeal** or

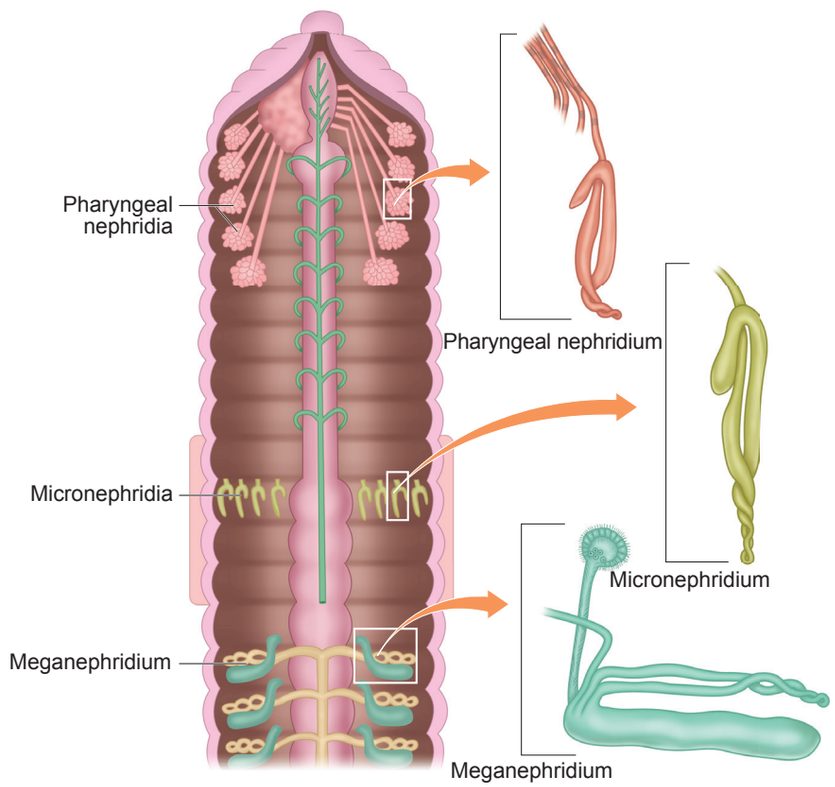


Figure: 4.5 *Lampito mauritii* – Types of Nephridia

TN_GOVT_XI_Zoology_Ch04.indd 52 09-01-2020 5.14.21 PM